

Supercharging Your L^AT_EX Workflow with Vim

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What is Vim?

- ▶ Vim is an open open sourced, terminal based modal text editor that is keyboard based, light-weight, and highly customizable.
- ▶ It is mainly used by software engineers to produce code or
- ▶ This means no mouse required!



What do you mean by Open Source?

Software that is freely available and may be redistributed and modified.

What is L^AT_EX?

- ▶ It is a high-quality typesetting system used for the production of technical and scientific documentation.
- ▶ It is the de-facto standard for communication and publication of scientific and scholarly articles.

What Do You Mean by Modal?

- ▶ Vim operates in different modes where each keystroke (or combination of keystrokes) executes different functions based on the current mode that you are in.
- ▶ There are three modes:
 - ▶ **Normal Mode (Command Mode):** This is the default mode of Vim. Keystrokes in this mode are meant for navigating around a file, manipulate text, and execute certain actions on text. Using the keys "hjkl" navigate the cursor around a file.
 - ▶ **Insert Mode:** This is the mode where you type in text into your file just as if you were typing on any other text-editor. To enable insert mode, you just have to "i".
 - ▶ **Command-Line Mode:** This mode allows you to execute more complex commands like saving files, quitting vim, and running external programs. To enter the command-line mode, press ":".

Examples of Common Commands

- ▶ Normal mode:
 - ▶ `"cc"` deletes a line and puts you into insert mode.
 - ▶ `"di("` deletes any text that is wrap around in parentheses.
 - ▶ `"ci["` deletes any text inside a set of brackets and puts you into insert mode.
- ▶ Command-line Mode:
 - ▶ `":w"` writes the file (save)
 - ▶ `":q"` quit vim
 - ▶ `":wq"` writes the file and quits out of vim.

Why Do I Care?

Vim being entirely keyboard based is perfect for optimizing the workflow of typesetting latex documents.

Problems Commonly Found in Modern Text-editors

- ▶ In most text-editors or IDE's, navigation revolves around the mouse which can be extremely inefficient.
- ▶ In IDE's, there are lots of potential for distraction due to the abundance of features available to you which may be overwhelming.
- ▶ Most programs have a large amount of "bloat" that will slow your program when latex files get large.

Problems Commonly Found Modern Text-Editors (Cont.)

- ▶ If you are using an online-based text editor, like overleaf, server crashes may halt your work and may even get rid of your work altogether.
- ▶ Having your work stored locally in your machine helps prevents any chance of losing your work and any frustration that may come from it.



What Are The Advantages?

- ▶ Vim is lightweight and does not require any sophisticated hardware to be able to run properly. It doesn't even require the internet to run!
- ▶ There is no "GUI" needed to access the editor. You just need to use the terminal.
- ▶ It is entirely local which may ease the fear of having your files tampered with if you are using an online-based editor.

What Are The Advantages (Cont.)

- ▶ It is barebones by design which means it is not distracting when you open it up initially. However, do not confuse this with not having a lot features! It does have a lot of it, you just have to uncover it yourself.
- ▶ Since it is entirely keyboard based, you do not have to lift your hand off the keyboard to navigate a file which is HUGE.

The Most Annoying Thing with $\text{\LaTeX}$

- ▶ With $\text{\LaTeX}$, all of the code produced involve typing " $\backslash$ " which is cumbersome to type out every single time.
- ▶ We can easily avoid this nuisance by creating "snippets" of all our commonly typed text.
- ▶ These snippets along with the keyboard centric workflow of vim makes for an extremely fast and efficient workflow.

Examples of Commonly Typed Text

- ▶ For inline math, type `\( \)`.
- ▶ Fractions, type `\frac{}{}`
- ▶ Kind of environments like `\begin{...} ... \end{...}`

What Is A Snippet?

- ▶ A snippet is a short reusable piece of text or code that can be triggered by entering a combination of keystrokes.
- ▶ These pieces of text are stored in a separate file that is executed every time you hit keystrokes corresponding to the text that you want expanded out.
- ▶ These snippets are triggered either manually or automatically.

Difference Between Autocomplete and Snippets

Structure Of A Snippet